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Repeated exposures to diisopropylfluorophosphate result in structural disruptions of myelinated axons and persistent impairments of axonal transport in the brains of rats

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ABSTRACT

Organophosphates (OPs) are found in hundreds of valuable agricultural, industrial, and commercial compounds; however, they have also been associated with a variety of harmful effects in humans. The acute toxicity of OPs is attributed to the inhibition of the enzyme acetylcholinesterase (AChE); however, this mechanism may not account for all of the deleterious neurologic effects of OPs, especially at doses that produce no overt signs of acute toxicity. In this study, the effects of two weeks of daily subcutaneous exposure to the OP-nerve agent diisopropylfluorophosphate (DFP) in doses ranging from 0.125-0.500 mg/kg on whole brain volume, white matter, and gray matter integrity were evaluated in post mortem tissues using histology and magnetic resonance imaging (MRI) methods. The effects of DFP on axonal transport in the brains of living rats were evaluated using a manganese-enhanced MRI (MEMRI) method. DFP was associated with dose-dependent impairments in red blood cell and brain AChE (down to 29 and 18% of control, respectively at the highest dose), 24 h after the last injection. However, there were no visible signs of cholinergic toxicity noted in any portion of the study. Moreover, histological and MRI analysis of post mortem brains did not reveal any pronounced alterations of whole brain, white matter, or gray matter volumes associated with DFP. Electron microscopy did reveal a DFP-related increase in structural disruptions of myelinated axons (i.e., decompactions) in the fimbria region on the corpus callosum. MEMRI indicated that DFP was also associated with dose-dependent decreases in axonal transport in the brains of living rats, an effect that was also present after a 30-day (DFP-free) washout period, when AChE was not significantly inhibited. These results indicate that repeated exposures to the nerve agent, DFP at doses that are below the threshold for acute toxicity, can result in alterations in myelin structure and persistent decreases in axonal transport in the rodent brain. These observations could explain some of the long-term neurological deficits that have been observed in humans who have been repeatedly exposed to OPs.

1. Introduction

The chemicals known as the "organophosphates" (OPs) are found in hundreds of useful agricultural, industrial, and commercial products including pesticides, defoliants, fire retardants, industrial solvents, lubricants, plasticizers, and fuel additives (reviewed, Soltaninejad and Shadnia, 2014). Unfortunately, many OPs are highly toxic and both accidental and intentional exposures to OPs resulting in deleterious health effects have been documented for decades (reviewed in Worek et al., 2016; Masson and Nachon, 2017). In the case of acute poisoning, a relatively sound causal argument can be made for the mechanism of both the acute symptoms of OP toxicity and the long-term neurological consequences. In this case, inhibition of acetylcholinesterase (AChE) by the OP leads to marked elevations in synaptic acetylcholine levels which in turn lead to excessive stimulation of both muscarinic and nicotinic acetylcholine receptors (Ecobichon, 2001, for review see also Pereira et al., 2014). Depending on the exposure level, some of the acute effects of OPs including excessive secretions, cardiorespiratory depression, and seizures can be life threatening, but they can also lead to a variety of long-term neurological and psychiatric consequences in survivors. These long-term effects of OPs can include motor impairments, psychiatric disturbances, and deficits in attention, learning, and memory (reviewed in Pereira et al., 2014).

In contrast to the consequences of acute OP poisoning, the

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mechanism of the long-term adverse effects of repeated exposures to levels of OPs below the threshold for acute "cholinergic" toxicity are somewhat less clear (see below). This type of exposure is relatively common in agricultural workers and pesticide sprayers and it has been associated with persistent alterations in psychomotor speed, executive function, visuospatial ability, working and visual memory (see Ross et al., 2013). Lower level, repeated exposures to OPs have also been associated with a variety of adverse symptoms in other contexts. For example, low-level exposures to OP-based insecticides (e.g., chlorpyrifos, dichlorvos) as well as nerve agent-OPs (sarin and cyclosarin) following the destruction of an Iraqi munitions storage complex at Khamisiyah, Iraq, in March 1991 have been implicated in the etiology of Gulf War Illness (GWI), which affects up to one-fourth of the veterans from the first gulf war. GWI is characterized by a complex set of symptoms including unexplained fatigue, respiratory difficulties, musculoskeletal pain, gastrointestinal distress, skin rashes, headaches, and a variety of neurological and neuropsychiatric problems including cognitive impairment (Research Advisory Committee on Gulf War Veterans' Illnesses, 2014). It is also worth noting that a recent study reported similar symptoms in Kurdish civilians who were exposed to chemical warfare agents (including OPs) as children, during the events leading up to the Gulf War (Talabani et al., 2017). In addition, repeated exposures to the OP, tri-cresyl phosphate (TCP), used as an anti-wear additive to jet engine oil, has been implicated in "Aerotoxic Syndrome" a term used to describe acute and persistent symptoms reported by aircrew following exposures to fumes in aircraft cabins. Symptoms include ear/nose/throat irritation, skin conditions, nausea and vomiting, respiratory problems, headaches, weakness and fatigue, nerve pain, tremors, and cognitive impairment (see review, Harrison and Mackenzie Ross, 2016; Michaelis et al., 2017).

While the symptoms associated with OPs in each of the scenarios described above (agricultural exposures, GWI, Aerotoxic Syndrome) can be quite diverse, neurologic impairments, particularly deficits in cognitive function are a commonly reported feature of each condition. Some neuroimaging studies have been conducted to identify possible neuroanatomical correlates of the long-term neurological and behavioral symptoms described above. For example, structural magnetic resonance imaging (MRI) has revealed reduced white and gray matter volumes in cortical areas in symptomatic Gulf War veterans and in sarin/cyclosarin-exposed Gulf War veterans (Chao et al., 2011, 2010; Rayhan et al., 2013; Rosenzweig et al., 2012). Functional MRI studies of Gulf War veterans have revealed alterations in multiple brain regions including the thalamus, caudate, hippocampus, globus pallidus and putamen (Calley et al., 2010; Haley et al., 2009; Li et al., 2011). Using diffusion tensor imaging and arterial spin labeling, preliminary evidence of white matter and cerebral perfusion alterations have also been reported in aircrew members exposed to engine oil fumes who self-reported cognitive deficits and depressive symptoms (Reneman et al., 2016).

It is important to note that there are a number of confounding factors that can limit the interpretations of the human studies described above, particularly when attempts are made to make causal connections between illness symptoms and specific underlying biological mechanisms. Most of the human literature on OP exposures is based on retrospective investigations, case reports, and epidemiology and in each of the conditions, the study subjects were likely exposed to multiple substances as well as additional environmental insults (e.g., heat, stress, smoke, high altitude). Other factors that could confound the interpretations of the aforementioned studies include the inability to establish clear dose-effect relationships and the length of the periods of exposure to particular OPs. Therefore, a critical need exists for prospective studies (that can only be conducted ethically in non-human model systems) to determine what neurobiological consequences can indeed be linked directly to repeated exposures to particular OPs and when adverse effects can be linked, to establish dose-effect and exposure-time relationships and well as the underlying biological mechanisms of the adverse effects.

Regarding neurobiological mechanisms of the toxic effects OPs, significant evidence suggests that it may not be fully explained by AChE inhibition (reviewed, Banks and Lein, 2012; Terry et al., 2012). Previous work in our laboratory suggests that OP-related impairments of axonal transport may contribute to the long-term alterations in neuronal function. Specifically we have observed impairments in axonal transport in cultured neurons in vitro associated with both the insecticide OP chlorpyrifos as well as the nerve agent diisopropylfluorophosphate (DFP) (Gao et al., 2016, 2017). We have also observed axonal transport deficits in vivo in living rats exposed to chlorpyrifos using manganese-enhanced magnetic resonance imaging (MEMRI) (Hernandez et al., 2015), but have not yet evaluated DFP with this method. This is an important determination to make since chlorpyrifos and DFP are quite different structurally and metabolically (i.e., chlorpyrifos is a phosphorothioate that requires metabolic conversion to an oxon to be potent as an AChE inhibitor while DFP is an alkylphosphate nerve agent with a very rapid action as an AChE inhibitor).

In the study described here, the effects of repeated exposures to DFP on whole brain volume as well as white and gray matter integrity were evaluated in the post mortem rat brain using histological and MRI methods. We also evaluated the effects of DFP on axonal transport in the living rodent brain using MEMRI.

2. Material and methods

2.1. Animals

Male albino Wistar rats (Envigo, Indianapolis, IN, USA) 2–3 months old were housed in pairs in a temperature controlled room (25 °C), maintained on a standard 12-h light/dark cycle with free access to food (Teklad Global Rodent Diet 2918, Envigo, Madison, WI, USA). Table 1 provides the details for the study cohorts, the numbers of animals tested per group, and the experiments conducted in each group. All procedures employed during this study were reviewed and approved by the Augusta University Institutional Animal Care and Use Committee and are consistent with AAALAC guidelines. Measures were taken to minimize pain and discomfort in accordance with the National Institutes of Health, Guide for the Care and Use of Laboratory Animals (NRC, 2011). Significant efforts were also made to minimize the total number of animals used while maintaining statistically valid group numbers.

2.2. Drug administration

Diisopropylfluorophosphate (DFP) (Sigma Aldrich (CAS 55-91–4, St. Louis, MO). was prepared fresh daily in peanut oil (Kroger® Pure Peanut Oil, obtained locally, Augusta, GA, USA) and administered subcutaneously (s.c.) 0.7 ml/kg body weight. The doses evaluated in this study were identified in previous studies (Stone et al., 2000; Terry et al., 2011) and operationally defined as doses not associated with acute signs of cholinergic toxicity (e.g., muscle fasciculation, seizures, diarrhea, excessive urination, salivation, etc., see reviews, Rusyniak and Nanagas, 2004; Sungurtekin et al., 2006). In all cases illustrated in Table 1, the individual test subjects were randomly assigned to a particular DFP dose or vehicle group.

After each daily injection, individual rats were monitored (in their home cages for a period of approximately 5 min.) for visible cholinergic signs (diarrhea, excessive salivation or lacrimation, respiratory difficulties, muscle fasciculation) or other signs of distress throughout the study. Subjects were also weighed at least twice per week and monitored for any signs of distress during the weighing period.

2.3. Acetylcholinesterase activity

Acetylcholinesterase activity was assessed in red blood cells (RBCs) and brain using the method of Ellman (Ellman et al., 1961) with

Table 1
Rat Experimental Protocol.

Cohort	Group	N	Treatment x 14 days	Days of Washout/Procedure Conducted		
				Baseline	1	30
1	A	6	Vehicle		AChE Activity	
	B	6	DFP 0.125 mg/kg/day		AChE Activity	
	C	6	DFP 0.250 mg/kg/day		AChE Activity	
	D	6	DFP 0.500 mg/kg/day		AChE Activity	
2	E	6	Vehicle			AChE Activity
	F	6	DFP 0.125 mg/kg/day			AChE Activity
	G	6	DFP 0.250 mg/kg/day			AChE Activity
	H	6	DFP 0.500 mg/kg/day			AChE Activity
3	I	6	Vehicle		Histology (CV)	
	J	6	DFP 0.125 mg/kg/day		Histology (CV)	
	K	6	DFP 0.250 mg/kg/day		Histology (CV)	
	L	6	DFP 0.500 mg/kg/day		Histology (CV)	
4	M	6	Vehicle		Histology (LFB)	
	N	6	DFP 0.125 mg/kg/day		Histology (LFB)	
	O	6	DFP 0.250 mg/kg/day		Histology (LFB)	
	P	6	DFP 0.500 mg/kg/day		Histology (LFB)	
5	Q	6	Vehicle		DTI	
	R	6	DFP 0.500 mg/kg/day		DTI	
6	S	4	Vehicle		EM	
	T	4	DFP 0.500 mg/kg/day		EM	
7	U	6	Vehicle	MEMRI	MEMRI	MEMRI
	V	6	DFP 0.125 mg/kg/day	MEMRI	MEMRI	MEMRI
	W	6	DFP 0.250 mg/kg/day	MEMRI	MEMRI	MEMRI
	X	6	DFP 0.500 mg/kg/day	MEMRI	MEMRI	MEMRI

Abbreviations, DFP = Diisopropylfluorophosphate, AChE, acetylcholinesterase, LFB, luxol fast blue, CV, cresyl violet, DTI, diffusion tensor imaging, EM, electron microscopy, MEMRI, manganese-enhanced magnetic resonance imaging.

modifications to accommodate a 96-well microplate format at 25 °C (see Terry et al., 2007). Blood and brains were collected in separate sets of animals in parallel with the animals in the histology and MRI studies (see Table 1) at two time points, 1) 24 h after completing 14 days of repeated DFP exposure (0.125-0.500 mg/kg, s.c.), and 2) 30 days after completing 14 days of repeated DFP exposure 0.125-0.500 mg/kg, s.c.). Subjects were anesthetized with isoflurane and transcardially perfused with ice-cold phosphate-buffered saline (PBS) to thoroughly clear the brain of blood, particularly peripheral borne butyrylcholinesterases. Red blood cells were collected in Potassium/EDTA Vacutainers (BD®, Franklin Lakes, NJ) and spun in a clinical centrifuge for 15 min. to separate RBC's and plasma. Plasma was aspirated and a 1:50 dilution containing 50 µl of RBC's and 2450 µl of 0.9% (w/v) blood bank saline with 1.0% Triton X-100 (Sigma, St. Louis, MO) was made. RBC's were frozen and stored at −80 °C until the Ellman assay was performed. Brains were extracted, rinsed with PBS and stored at −80 °C until use. Brain tissue was homogenized in ice-cold PBS (wt/vol: 1 *g* wet brain tissue/4 ml PBS) using a motor driven glass-Teflon tissue grinder. Brain and RBC total protein concentration was measured using a Micro BCA Protein Assay Kit (ThermoFisher Scientific Inc., Rockford, IL, USA) according to manufacturer's instructions. Brain homogenates (20–50 µg protein/µl) and RBC's (7–12 µg protein/µl) were assayed in triplicate for AChE using 0.48 mM acetylthiocholine (substrate) and 0.52 mM dithiobisnitrobenzoic acid diluted in 1.0 mM sodium phosphate buffer (see Terry et al., 2007). The reaction mixture for RBC's included 0.097 mM ISO-OMPA (Sigma, St. Louis, MO) and was pre-incubated with the samples five minutes prior to substrate addition for butyrylcholinesterase inhibition. The formation of reaction product (yellow color) was monitored by measuring absorbance values (in optical density) at 412 nm every 2 min. for 16 min (µQuant™ Microplate Spectrophotometer, BioTek Instruments Inc., Winooski, VT, USA). The cholinesterase-mediated reaction rate (moles/L per min) was calculated by dividing the change in absorbance per min by 13,600 (Ellman et al., 1961).

2.4. Histology

2.4.1. Tissue preparation

After repeated OP exposure, rats were anesthetized with isoflurane and euthanized by transcardial perfusion with phosphate-buffered saline (PBS) (pH 7.4), followed by fixative (4% (wt/vol) paraformaldehyde in PBS). Extracted whole brains were post-fixed overnight and cryoprotected (in 30% (wt/vol) sucrose for long-term storage at 4 °C. Cryosectioning was performed at the Histology Core Facility at Augusta University to obtain whole 40 µm sagittal sections starting at the midline. All sections were collected sequentially and stored at −20 °C in ethylene glycol-based cryoprotectant. For stereological analysis, every 5th section (200 µm between sections) was mounted in sequence onto charged microslides. To identify boundaries of target regions of interest (ROIs), cresyl violet and luxol fast blue stains were utilized to stain neuropil (neurons and glia) and myelin, respectively. Sections were adhered to slides with acetone then defatted with xylene and rehydrated with ethanol to stain with 0.5% (wt/vol) cresyl violet or 0.1% (wt/vol) luxol fast blue (both in acetate buffer). Following staining, slides were differentiated and clarified in ethanol and xylene, respectively, to coverslip with mounting media.

2.4.2. Stereological analyses

A semiautomatic stereology system (Stereologer, Stereological Resource Center, Baltimore, MD) was used by a trained individual blind to the treatment of each rat to measure the volume of these ROIs: hippocampus proper, hippocampus subregions and corpus callosum based on the Cavalieri principle (Bonthius et al., 2004). Briefly, anatomical landmarks were localized for each target ROI under low magnification (5X) to manually outline and a randomly placed grid was superimposed for automated point counting (area per point = 0.1 or 0.5 mm^2). Brain volumes were estimated by the Cavalieri method using the formula $V = \Sigma A^* T$, where ΣA is the surface area of each section and T is the distance between sections. The results for one hemisphere were multiplied by two to give whole brain volume estimations.

2.5. Magnetic resonance imaging (MRI) – post-mortem volume

All MRI data acquisition was performed at the Core Imaging Facility for Small Animals (CIFSA), Augusta University.

2.5.1. Image acquisition

Volumetric MRI analysis of postmortem rat brain was performed using modifications of a previously published protocol for mice (Howell and Pillai, 2014). The fixed brains were imaged with a Bruker BioSpec 7.0 T/ 20 cm horizontal bore MRI system (Bruker Instruments, Billerica, MA, USA). Images were acquired using a standard transmit/receive volume coil (35 mm i.d.) with the fixed rat brain in prone position. A Rapid Acquisition with Relaxation Enhancement (RARE) sequence was used to acquire anatomical images with contrast dominated by proton density to perform the volumetric measurement via image segmentation. Parameters of the scan were as follows: TE/TR = 9.0/1740 ms; matrix = 256 × 256; RARE factor = 4; field of view (FOV) = 2.0 × 2.0 cm^2; slice thickness = 0.75 mm; zero-filled acceleration = 1.33; number of averages = 6; and total scan time = 8 min 21 s.

2.5.2. Volume measurements

From ex vivo MR images, the volumes of the whole brain, cerebral cortex, and subcortical structures (striatum, hippocampus, lateral ventricles, and corpus callosum) were determined by two individuals who were blind to the group assignments using a similar method to that described by Vernon et al. (2011). For each brain region, tracings were drawn on a slice-by-slice basis in the coronal plane (rostral to caudal) using the ROI tool in ImageJ software (National Institutes of Health, Bethesda, Maryland). For each structure, ROI contours were traced in both brain hemispheres and combined. Sample ROI contours for each region from a vehicle-control rat are shown in Fig. 3. Clear anatomical landmarks referenced to a standard rodent brain atlas (Paxinos and Watson, 1998) were used to define ROI contours. The following anatomical criteria were used for the measurement of each brain structure: whole brain, 36 consecutive slices from the olfactory bulb to the end of the cerebellum (approx. +6.70 to −14.60 mm from bregma), cerebral cortex, 17–19 consecutive slices (approx. +3.70 to -8.00 mm from bregma), defined with reference to corpus callosum, external capsule and rhinal fissure, corpus striatum, 12–14 consecutive slices (approx. +2.20 to -1.60 mm from bregma), defined with reference to corpus callosum, external capsule, anterior commissure and lateral ventricles, hippocampus, 9–11 consecutive slices (approx. -1.88 to -6.04 mm from bregma), defined with reference to corpus callosum and external capsule, corpus callosum, 7–9 consecutive slices (approx. + 1.60 to -5.30 mm from bregma), defined with reference to cingulum and external capsule, lateral ventricles, 6–8 consecutive slices (approx. + 1.20 to -0.80 mm from bregma), defined from brain tissue by intense contrast of CSF. Volumes were calculated by multiplying the sum of the areas measured for each brain structure (corrected for the different number of slices visualized in each image stack) by the slice thickness (750 μm).

2.6. Diffusion tensor imaging (DTI)

For DTI analysis of the corpus callosum in post mortem brains, a diffusion-weighted spin echo sequence was performed on the same 7 T MRI system, with the following parameters: TE/TR = 18.8/4000 ms; number of averages = 6; number of diffusion directions = 12; number of b0 images (without diffusion gradient) = 2; b value = 1700 s/mm^2; acquisition matrix = 128 × 128; number of slices = 16; slice thickness = 1.5 mm; FOV = 2.0 × 2.0 cm^2; and total scan time = 11 h 57 min. The raw gradient-direction-dependent diffusion-weighted images were then processed using Bruker's Paravision 5.1 software to reconstruct DTI map images for all slices that were further respectively categorized into the fractional anisotropy (FA), trace of diffusion tensor, b0 image, trace weighted image, diffusion tensor component, eigenvalue and eigenvector, respectively, using ImageJ (Abramoff et al., 2004) with an 'in-house' scripted plug-in. Specifically, we manually outlined the corpus callosum (a prevalent myelinated structure) for FA parametric maps. Raw FA intensity values were corrected for the area measured (in pixel number). Mean FA values for the corpus callosum are reported.

2.7. Electron microscopy (EM) experiments

We also added electron microscopy (EM) as a method to analyze axons at a higher level of resolution to determine if demyelination and/or myelin-related damage was associated with DFP exposure. EM analysis of myelin was performed using a modification of the procedure described previously by Tao et al. (2009). Rats were anesthetized and cardiac perfused with 4% formaldehyde and 2% glutaraldehyde in 0.1 M sodium cacodylate buffer (NaCac; pH 7.4). Brains were removed and fixed overnight at 4 °C in the same perfusion fixative for 24 h. They were washed by 0.1 M NaCac and osmicated with 2% osmium tetroxide for 30–60 min at 4 °C, washed by 0.1 M NaCac and by deionized H_2O for 30–60 min at 4 °C, and dehydrated in graded (30–70%) ethanol. Whole brains were sectioned at 500 microns in the coronal plane to isolate the fimbria, a region connecting the corpus callosum to the hippocampus. The fimbria was dissected and processed to stain with 2% uranyl acetate in 70% ethanol at 4 °C for 30 min., followed by dehydration with 70–100% ethanol. Samples were incubated two times with propylene oxide for 10 min. and embedded with resins. Ultrathin sections (65 nm) were collected using an ultramicrotome (Reichert Ultracut T and Leica UC6) from resin-embedded fimbria, oriented to obtain a cross-sectional view of axons.

EM micrographs collected at 15,000 X magnification were digitized with a JEM 1230 Transmission Electron Microscope and converted to TIFF format to assess axon morphology. Briefly, at the FOV in which analysis was conducted, axons were sorted into two categories based on the intensity of electron density staining surrounding each: myelinated (black or dark) or unmyelinated (unstained or light) (Payne et al., 2011) (See Fig. 5). Axons were classified as "myelinated" when up to a minimum of 85% of the exterior surface was bordered with dense electrons (i.e., compact myelin). Some axonal surfaces exhibited more than 15% disruption to the interaction between the multiple layers of lamellae (between axon and myelin) and classified as "decompacted myelin." Axons were categorized as "unmyelinated' in the absence of an electron dense surface surrounding the exterior of an axon. Additionally, partial axons were excluded if more than 15% of their surface area was outside the boundaries of the micrograph. Based on these select criteria, myelin integrity in the fimbria portion of the corpus callosum was evaluated following exposure to DFP in comparison to its corresponding vehicle treatment (18–25 images per rat, n = 4 rats per group). For quantitation, the total number of myelinated and unmyelinated axons per field of view as well as g-ratios and the number of decompacted axons per field of view were counted manually by investigators who were blinded to the group identity.

In order to eliminate any potential bias based on circularity, g-ratios were calculated by the perimeter of the axons (inner) divided by the perimeter of the corresponding myelin sheath (outer) as described in Tao et al. (2009). All measurements were completed in NIH ImageJ.

2.8. Manganese-enhanced magnetic resonance imaging (MEMRI)

MEMRI studies were conducted as we have described in detail previously (see Hernandez et al., 2015). Below we have provided an abbreviated overview of the methods.

2.8.1. Intravitreal manganese administration

Rats were anesthetized with a mixture of ketamine, 100 mg/kg i.p. and xylazine, 10 mg/kg i.p. prior to each intravitreal injection. $MnCl_2$ (200 μM/4 μL) was injected with a 30G1½ needle (Beckon-Dickinson

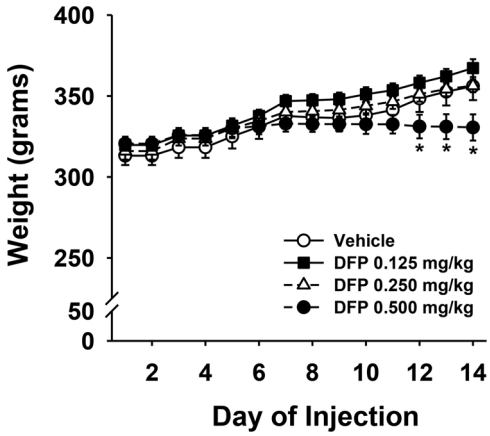

Fig. 1. Effect of daily subcutaneous injections of vehicle or DFP on body weight over a 14-day period. Each point represents the mean ± S.E.M. N = 12 rats/group. *p < 0.05, compared with vehicle.

Hypodermic #306106) behind the lens to access the vitreous humor of the left eye using care to avoid reflux after removal. Animals were returned to their home cage to be monitored for signs of distress and fully recover prior to each MRI session.

2.8.2. MRI

MRI scans were collected 6 and 24 h following each intravitreal Mn^{2+} injection in three separate sessions at the following time points: 1) a pre-treatment baseline, 2) at the end of 14 days of DFP exposure, and 3) following a 30-day DFP-free washout period. Prior to and throughout the MRI imaging session, rats were anesthetized with a mixture of medical air, oxygen (1:1), and 2.5% isoflurane. Once anesthetized, the head was secured (using medical tape) to a thermo-controlled (37.8 °C) cradle while the remainder of the body was unrestrained to promote unrestricted respiration. ECG and respiratory signals were monitored by a physiological monitoring system (SA Instruments, INC., Stony Brook, NY). Initial imaging using a three-plane, 2D T1W Fast Low Angle Shot (FLASH) sequence (TE/TR = 3.6/100 msec; Matrix = 128 × 128; FOV = 3.84 × 3.84cm^2; flip angle = 30°; and three 0.5-mm thick tri-orthogonal slices) was performed to confirm the successful injection of Mn^{2+} into the eyeball. Visualization of the optic nerve enhancement was realized using a 3D FLASH sequence (TE/TR = 3.7/12.5 ms; number of averages = 10; Matrix = 192 × 192 × 192; field of view = 3.84 × 3.84 × 3.84 cm^3; and flip angle = 30°). Total scan time was approximately 60 min with total time in the scanner from start to finish being approximately 70 min.

2.8.3. MEMRI data analysis

Based on our previously published study (Hernandez et al., 2015), we determined that analysis of the superior colliculus (SC) 24 h after intravitreal injection of $MnCl_2$ was the best approach for making comparisons of axonal transport between treatment groups. In the current study, the 6-hour time point was used primarily to confirm successful injection of Mn^{2+} into the eyeball and that the contrast agent was progressing along the optic nerve as expected. Each MRI image set was imported as a raw data file (16-bit, 192 × 192 × 192 pixels) and analyzed using a semi-automated method with ImageJ software (Abramoff et al., 2004). Mn^{2+}-enhancement in the SC was localized and identified with reference to the Rat Brain Atlas (Paxinos and Watson, 4th Edition) to establish neuroanatomical boundaries. Both Mn^{2+} enhanced and contralateral (non-enhanced) SC were manually outlined and analyzed to obtain a mean intensity value, defined as the sum of each voxel's raw intensity value contained with an outline divided by the total number of voxels. For each image, the enhanced SC intensity value was divided by the contralateral non-enhanced SC intensity value to obtain a normalized SC intensity value. In turn, the mean normalized SC intensity was obtained from all image slices in which the SC appeared (10–11 per subject) within each time point for each subject.

2.9. Statistical analyses

All statistical analyses were performed using SigmaPlot 11.2 (Systat Software, San Jose, CA). Student's t-tests or analysis of variance (with repeated measures when indicated) were used to compare dose (vehicle vs. DFP at different doses) and time (baseline (BL), washout 1 (WO1) or washout 2 (WO2)) for the MEMRI experiments. Student Newman-Keuls multiple comparison procedures were used to examine post hoc differences when indicated. Statistical significance was assessed using an alpha level of 0.05. For each figure and table presented, error values denoted by ± indicate the standard error of the mean.

3. Results

3.1. Body weights visual observations

All test subjects were weighed upon arrival, then again daily during the DFP exposure period. Fig. 1 illustrates the body weights of representative test subjects (N = 12) from cohorts 1–4 (see Table 1) over the course of the 14-day DFP treatment period. Both vehicle and DFP-treated subjects gained weight as expected over the course of the two weeks, effect of weighing day, $F_{13,572} = 7.87$, $p < 0.001$. Statistical analysis did indicate a significant dose x day interaction. $F_{39,572} = 128.34$, $p < 0.001$, and post hoc analysis indicated that the DFP 0.500 mg/kg-treated subjects gained slightly, but significantly ($p < 0.05$) less weight (approximately 10 versus 30 g of weight gained) than the other three experimental groups on days 12–14 of treatment. As noted in the Methods individual test subjects were monitored daily for visible cholinergic symptoms and/or other signs of distress throughout the study. There were no cases where such signs of distress or cholinergic signs were detected in any portion of the study.

3.2. Acetylcholinesterase (AChE) activity

The effects of repeated exposures to DFP on RBC and brain AChE activity are provided in Table 2. AChE activity in RBCs was reduced (24 h after the last injection) in DFP-treated rats (effect of dose, $F_{3,27} = 18.6$, $p < 0.001$) to approximately 54%, 35%, and 29% of control in the 0.125, 0.250, and 0.500 mg/kg DFP groups, respectively (post hoc $p < 0.001$ for all 3 doses). After the 30-day DFP-free washout period, AChE activity in RBCs had fully returned to control levels in the lower dose group, whereas it was slightly (but not significantly) reduced in the higher dose groups (effect of dose, $F_{3,20} = 0.52$, $p > 0.6$). Similarly, AChE activity in brain was reduced (24 h after the last injection) in DFP-treated rats (effect of dose, $F_{3,22} = 92.5$, $p < 0.001$) to approximately 42%, 23%, and 18% of control in the 0.125, 0.250, and 0.500 mg/kg DFP groups, respectively (post hoc $p < 0.001$ for all 3 doses). After the 30-day DFP-free washout period, AChE activity in brain had fully returned to control levels in the lower dose group, whereas it was slightly (but again not significantly) reduced in the higher dose groups (effect of dose, $F_{3,20} = 2.98$, $p > 0.05$).

3.3. Histology

The results of histological experiments are presented in Fig. 2. All histological analyses were conducted on the brains of subjects that were sacrificed 24 h after the last DFP or vehicle injection. Quantitative stereological techniques were used to evaluate potential (DFP-related) alterations in volume of the hippocampal CA1 and CA3 pyramidal layers and DG granule cell layer as well as the volume of the whole hippocampus. Stereology was also used to evaluate potential DFP-

Table 2
Acetylcholinesterase Activity.

Treatment	Washout 1	% of Control	Washout 2	% of Control
		Red Blood Cells		
Vehicle	1.87 ± 0.24		1.34 ± 0.10	
DFP 0.125 mg/kg	0.98 ± 0.05**	54.54 ± 2.72	1.33 ± 0.08	99.32 ± 5.98
DFP 0.250 mg/kg	0.65 ± 0.06**	35.01 ± 3.00	1.22 ± 0.09	90.68 ± 6.34
DFP 0.500 mg/kg	0.55 ± 0.04**	29.20 ± 2.10	1.20 ± 0.14	89.64 ± 10.14
		Brain		
Vehicle	24.77 ± 1.51		18.50 ± 0.99	
DFP 0.125 mg/kg	10.45 ± 0.99**	42.13 ± 3.99	18.86 ± 0.89	101.94 ± 4.80
DFP 0.250 mg/kg	5.81 ± 0.50**	23.41 ± 2.02	16.98 ± 1.07	91.80 ± 5.79
DFP 0.500 mg/kg	4.39 ± 0.48**	17.72 ± 1.92	15.22 ± 0.88	82.28 ± 4.77

Acetylcholinesterase activity is expressed as nmoles acetylthiocholine hydrolyzed/min/mg protein.
Abbreviations, DFP = Diisopropylfluorophosphate; Washout 1 = 24 h after the last DFP or vehicle injection; Washout 2 = 30 days after the last DFP or vehicle injection. "**", $p < 0.001$ versus vehicle at the time point indicated. N = 6 for each determination.

related alterations of the corpus callosum. A representative cresyl violet-stained image of the hippocampus is provided in Fig. 2A and the mean volumes of the hippocampal subregions as well as the whole hippocampus associated with vehicle and each dose of DFP are presented as a histogram in Fig. 2B. There were no statistically significant differences in the volumes of the hippocampus associated with any of the doses of DFP compared to vehicle (all p values were > 0.05). A representative luxol fast blue stained image of the corpus callosum is provided in Fig. 2C and a histogram of the volumes associated with vehicle and each dose of DFP are presented as a histogram in Fig. 2D. Again, there were no statistically significant differences in the volumes of the corpus callosum associated with any of the doses of DFP compared to vehicle (all p values were > 0.05).

3.4. Brain volume measurements through anatomical MR images

All volume analyses were conducted on the postmortem brains of subjects that were sacrificed 24 h after the last DFP or vehicle injection. As noted in the *Methods* section, sample ROI contours for each brain region in anatomical MR images obtained ex vivo from a vehicle-control rat are shown in Fig. 3. The mean volumes of the whole brain, cerebral cortex, and subcortical structures (striatum, hippocampus, lateral ventricles, and corpus callosum) obtained ex vivo from vehicle and DFP (0.5 mg/kg) treated subjects are presented in Table 3. Statistical analysis indicated no significant differences in the volumes of any of the brain regions associated with DFP compared to vehicle (all p values were > 0.05).

3.5. Diffusion tensor imaging (DTI)

All DTI analyses were conducted on the postmortem brains of subjects that were sacrificed 24 h after the last DFP or vehicle injection. In white matter, axonal tracts favor water diffusing parallel to the long axes, but restrain water movement perpendicular to the axonal direction due to the longitudinal structure within the tissue. Therefore, through visualizing water diffusion, DTI can provide a quantitative approach to studying the microstructural components of white matter, including myelination and axon organization that complements the structural MRI studies of white matter volume (Asato et al., 2010). The results of the DTI analysis (Fractional Anisotropy-FA) of the corpus callosum from the post mortem brains of DFP (0.5 mg/kg)-treated rats compared to vehicle are illustrated in Fig. 4. A coronal slice of the rat brain atlas with detailed anatomical delineation and an FA parametric map from a representative subject treated with vehicle are provided in Fig. 4A while a histogram of the FA values associated with vehicle and DFP 0.5 mg/kg are presented in Fig. 4B. Statistical analysis indicated no

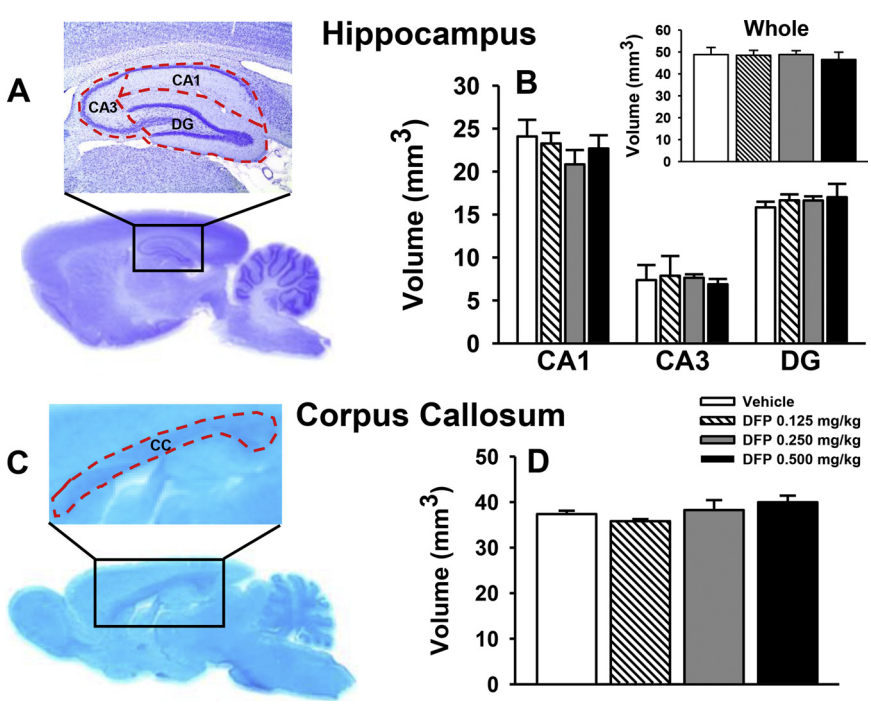

Fig. 2. Effects of 14-days of repeated exposures to DFP on the volume of the hippocampus and the corpus callosum (postmortem histology). Subjects were sacrificed 24 h after the last DFP or vehicle injection and the brains were prepared for histology (see Methods). (A) Representative histological image (sagittal plane) of the hippocampal formation from sections treated with cresyl-violet. The enlarged image illustrates the hippocampal subregions that were measured: CA1 pyramidal, CA3 pyramidal, and DG granule neuron layers. (B) Histogram illustrating the effects of the different doses of DFP compared to vehicle on total hippocampal volume and (C) on the volumes of different subregions of the hippocampus. (D) Representative histological image of the corpus callosum (sagittal plane) from sections treated with luxol fast blue. (E) Histogram illustrating the effects of the different doses of DFP compared to vehicle on the volume of the corpus callosum. The Cavalieri principle (point counting) was employed to measure volumes using unbiased stereological methods. N = 6 rats per group.

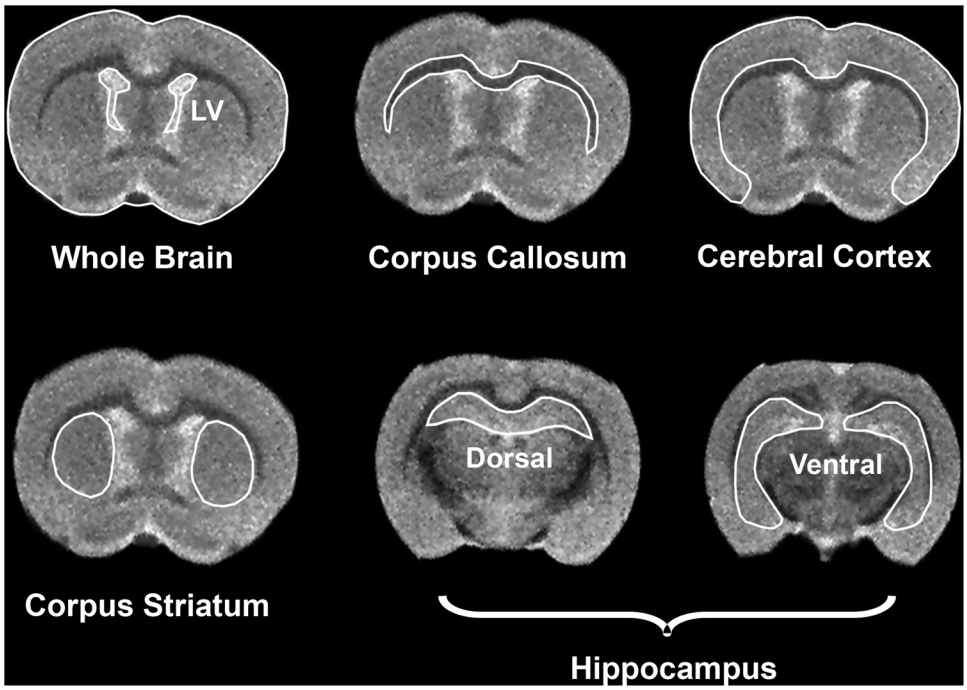

Fig. 3. Representative MRI coronal images with contrast dominated by proton density acquired ex vivo from a vehicle-treated rat to illustrate the brain regions of interest manually outlined for the volumetric determinations shown in Table 3. Contour borders were defined as described in the methods using a standard rodent brain atlas. LV = lateral ventricles. The post mortem images shown in this figure and the data shown in Table3 were acquired from experimental subjects that were sacrificed 24 h after the last DFP or vehicle injection (see Methods).

Table 3
Volumetric Data for Individual Brain Region from Ex Vivo MRI Scans (mm^3 ± sem).

	Whole Brain	Hippocampus	Striatum	Cortex	LV	CC
Vehicle	1819.85 ± 46.81	81.58 ± 3.10	71.22 ± 2.54	428.51 ± 11.81	5.16 ± 0.53	53.28 ± 1.94
DFP 0.5 mg/kg	1848.99 ± 63.78	90.26 ± 6.10	71.31 ± 2.30	413.12 ± 13.19	4.71 ± 0.32	53.06 ± 1.98

Abbreviations, DFP = Diisopropylfluorophosphate, LV = lateral ventricles, CC = corpus callosum. N = 6 for each determination.

significant differences in FA values associated with DFP compared to vehicle ($p > 0.05$).

3.6. Electron microscopy (EM) experiments

All EM analyses were conducted on the postmortem brains of subjects that were sacrificed 24 h after the last DFP or vehicle injection. Fig. 5 illustrates the effects of DFP 0.5 mg/kg on axons in the fimbria region of the corpus callosum. Sagittal and coronal rat brain maps illustrating the brain slice level, respectively, where transmission EM images were acquired are shown in Fig. 5A, representative images from a vehicle and DFP-treated rat are illustrated in Fig. 5B, and representative images demonstrating the categories for scoring 3 types of myelination are illustrated in Fig. 5C. Histogram plots indicating the total number of myelinated and unmyelinated axons per field of view as well as g-ratios and the number of decompacted axons per field of view

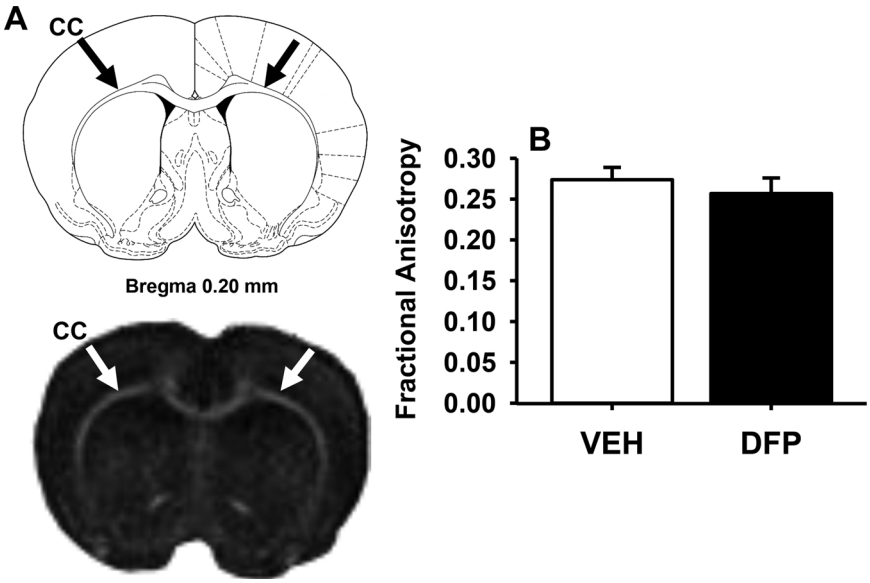

Fig. 4. Effects of 14-days of repeated exposures to DFP on the corpus callosum (postmortem MRI-diffusion tensor imaging). Subjects were sacrificed 24 h after the last DFP or vehicle injection and the brains were prepared for MRI analysis (see Methods) (A) Top-Representative coronal rat brain map (edited from the 4th edition of Paxinos and Watson Rat Brain Atlas) and Bottom-an FA parametric map from subject treated with vehicle. cc = corpus callosum (B) Histogram illustrating the effects of the DFP (0.5 mg/kg) compared to vehicle on FA. N = 6 rats per group.

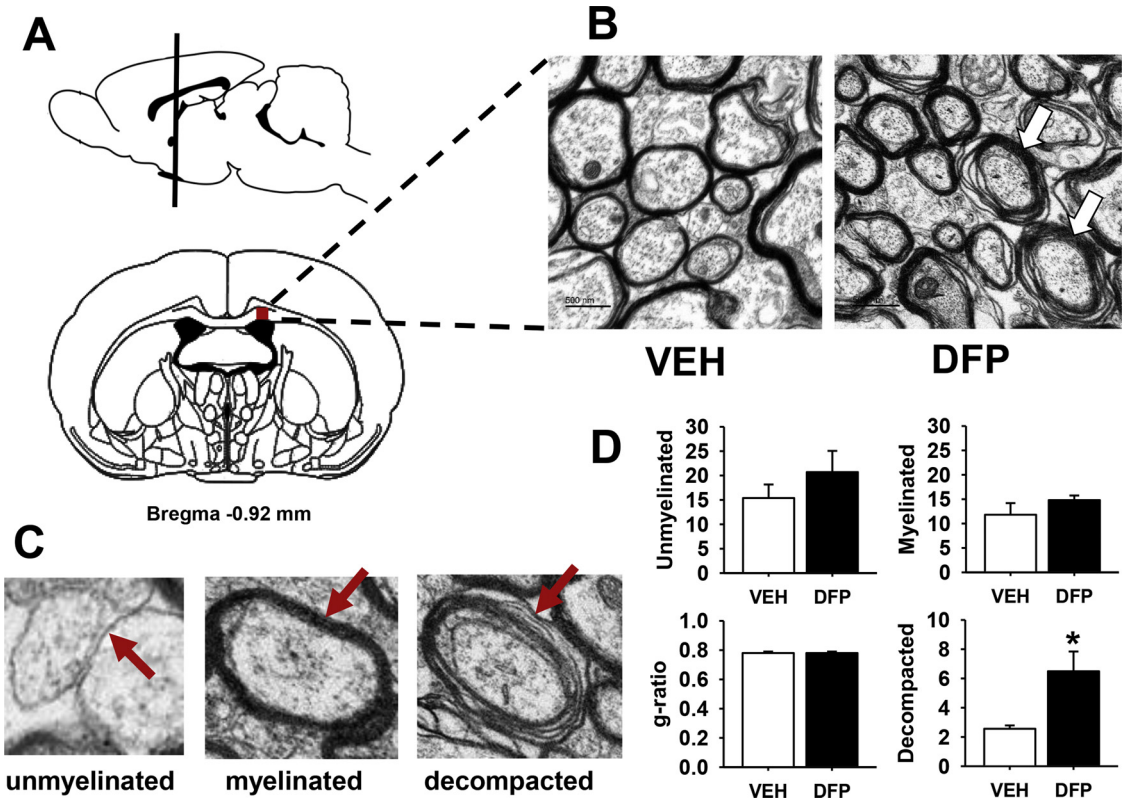

Fig. 5. Effects of 14-days of repeated exposures to DFP on axon morphology within the fimbria region of the corpus callosum (postmortem electron microscopy studies). Subjects were sacrificed 24 h after the last DFP or vehicle injection and the brains were prepared for EM analysis (see Methods) (A) Sagittal and coronal rat brain map (edited from the 4th edition of Paxinos and Watson Rat Brain Atlas) illustrating the brain slice level where transmission electron microscopy (TEM) images were acquired. The filled red square indicates the region isolated for cross-sectional analysis of the corpus callosum at the level of the fimbria. (B) Representative TEM images from a vehicle and DFP-treated rat. Scale Bar = 500 nm. (C) Representative TEM images demonstrating categories for scoring 3 types of myelination based on electron density (see red arrows): unmyelinated (low electron density), normally-myelinated (high electron density), and abnormally myelinated (decompacted) axons. (D) Histogram plots indicating the total number of myelinated and unmyelinated axons per field of view as well as g-ratios and the number of decompacted axons per field of view. *p < 0.05, compared with vehicle. N = 4 rats per group.

are provided in Fig. 5D. There were no statistically significant differences in the total number of myelinated or unmyelinated axons or in the g-ratios in DFP versus vehicle study groups (all p values were > 0.05). In our analysis, g-ratios typically ranged between 0.70 and 0.85 which is similar to the range that is considered normal for rat corpus callosum of 0.75–0.81 (see review, Chomiak and Hu, 2009). Here it is important to note that g-ratios were calculated in normally-myelinated axons only, as abnormally-myelinated axons often possessed irregular sheaths of myelin that varied in thickness around the axon and their inclusion would artificially skew the data. In contrast, there was a clear increase in the numbers of decompacted axons in DFP-treated rats versus vehicle-treated controls (p = 0.028, DFP versus vehicle-treated subjects).

3.7. Manganese-enhanced magnetic resonance imaging (MEMRI)

The effects of repeated exposure to DFP compared to vehicle administered by subcutaneous injection (once daily for 14 days) on axonal transport within retino-collicular projections of living rats are illustrated in Fig. 6. Fig. 6A illustrates the intravitreal injection method used in this study and the transport of Mn^{2+} within the axons of the optic nerve pathway. Histograms illustrating the normalized intensity values in the superior colliculus of vehicle and DFP-treated animals are provided in Fig. 6B and representative rat brain horizontal MR images from the vehicle and DFP treated groups are provided in Fig. 6C. Statistical analysis of the normalized intensity values in the superior colliculus, 24 h after intravitreal Mn^{2+} injection revealed the following, group (dose) effect, $F_{(3,20)} = 3.13$, $p = 0.047$, effect of the time period (i.e., duration of time following 14-days of DFP exposure), $F_{(2,33)} = 3.83$, $p = 0.032$; group x time period interaction, $F_{(6,33)} = 2.75$, $p = 0.028$. Post hoc analysis indicated that signal intensity was significantly ($p < 0.05$) higher at WO2 compared to BL and WO1 for the lowest dose of DFP. In contrast, signal intensity was significantly lower ($p < 0.05$) at WO1 and WO2 for the 0.5 mg/kg DFP group compared to vehicle-treated controls. In addition, within group comparisons indicated that DFP 0.5 mg/kg, but not 0.125 or 0.250 mg/kg was associated with reduced Mn^{2+} enhancement ($p < 0.05$) in the superior colliculus at both WO1 and WO2 compared to the respective baseline values.

4. Discussion

The major findings of this study can be summarized as follows: 1) Histological and MRI analysis of post mortem rat brains did not reveal any pronounced DFP-related alterations of whole brain, white matter, or gray matter volumes, 2) electron microscopy did reveal a DFP-related increase in structural disruptions of myelinated axons (i.e., decompactions) in the fimbria region of the corpus callosum, and 3) MEMRI indicated that DFP was also associated with dose-dependent decreases in axonal transport in the brains of living rats, an effect that was also present after a 30-day (DFP-free) washout period.

Given the previous reports of both structural and functional alterations in white and gray matter in neuroimaging studies in Gulf War veterans (some of whom were known to have been exposed to sarin/cyclosarin), it was our original hypothesis that we would observe similar effects in the brains of DFP-exposed rodents. While the decompactions in myelin and axonal transport deficits would be considered as

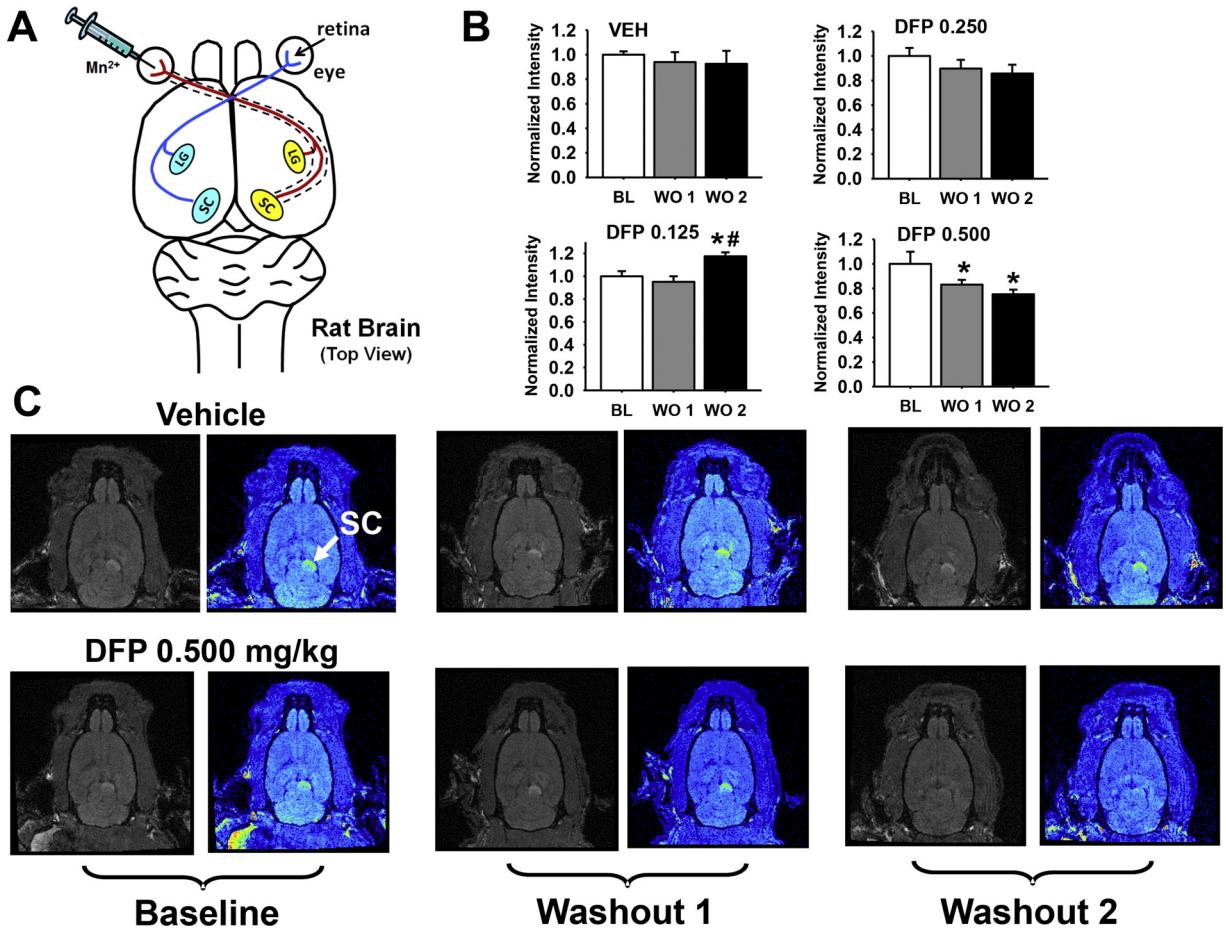

Fig. 6. Effects of repeated exposures to DFP on axonal transport in the brains of living rats. (A) Diagram illustrating the intravitreal injection method used in this study and the transport of Mn^{2+} within the axons of retinal ganglion cells (retina) in the anterograde direction (red line) along the optic nerve to the contralateral superior colliculus (SC) and lateral geniculate nucleus (LGN). (B) Histograms illustrating the normalized intensity values (mean ± s.e.m.) in the superior colliculus of vehicle and DFP-treated animals. Baseline = before repeated DFP or vehicle exposures, Washout 1 = one day (24 h) after the last DFP injection; Washout 2 = 30 days after the last DFP injection. * = significantly lower ($p < 0.05$) than baseline value. * = significantly higher ($p < 0.05$) than baseline value. (C) Representative rat brain horizontal MR images (approximately interaural 6.14 mm, bregma -3.86 mm) from the vehicle and DFP 0.500 mg/kg treated groups. Both the original black and white and color-enhanced images are provided. Note the clear Mn^{2+} enhancement in the superior colliculus (indicated by white arrow) of all groups at baseline and reduced enhancement associated with DFP (0.500 mg/kg) at washout 1 and 2. N = 6-7.

structural and functional alterations, respectively, our white and gray matter-related neuroimaging (MRI) and histological analyses were negative. Several potential explanations for these observations could be offered, but most would relate to challenges in modelling exposures/conditions in gulf war veterans. These challenges would include the unknown OP dose and contact periods, potential exposures to multiple substances (e.g., vaccines, prophylactic drugs against nerve agent exposures) and additional environmental insults (e.g., heat, stress, smoke). Another potential factor could be the long period between the OP exposure and the time the neuroimaging tests were performed in the veterans.

The DFP-related disruptions of myelin sheaths (decompactions) we observed in the fimbria region of the corpus callosum in the electron microscopy experiments were interesting. Myelin sheaths are projections of the plasma membrane of oligodendrocytes that wrap axons and myelin layers and when they are compacted, they increase axon conduction velocity (Bauer et al., 2009; Franklin and Gallo, 2014). Decompactions of myelin sheaths often accompany axon swellings subsequent to neuron injury and they are often sustained throughout chronic secondary degeneration. We did not observe axon swellings or other evidence of a more severe acute injury. It should be noted, however, that CNS myelin is rich in lipids (e.g., phospholipids) which are highly vulnerable to peroxidation. Lipid peroxidation is a known pathophysiological event occurring during secondary degeneration following nerve injury, and a variety of self-perpetuating events including the failure of mitochondrial functioning and antioxidant mechanisms result in elevations of reactive radical species that react with cellular membranes, including myelin (see Payne et al., 2012). Notably, it is well documented that OPs can increase oxidative stress as well as impair mitochondrial function (reviewed, Soltaninejad and Abdollahi, 2009; Banks and Lein, 2012; Terry, 2012).

The effects of DFP on myelin sheaths in the fimbria region may be especially relevant to previous observations of DFP-related memory impairments. The fimbria contains commissural fibers that project via the ventral hippocampal commissure to the CA1, CA2 and CA3 regions of the hippocampus and it also contains fibers that project via the fornix to multiple brain regions (Andersen, 2007). Importantly, hippocampal involvement in memory function has been repeatedly demonstrated through lesions of the fimbria–fornix, which normally relays hippocampal output to the septum (Levin et al., 1999; Balse et al., 1999; Levin et al., 1993). Moreover, exposure to DFP and other OPs (e.g., chlorpyrifos) have been associated with prolonged impairments of spatial learning tasks (e.g., the Morris water maze) which are dependent on intact hippocampal function (Terry et al., 2012).

Our observations of DFP-related decompactions at the high-resolution microscopic (EM) level without obvious effects in the lower

resolution histological and MRI anatomical images may be explained by the lack of significant changes in the total number of myelinated and unmyelinated axons and/or the g-ratios. This would indicate that the DFP dosing regimen influences the axonal structure microscopically, but does result in overt neuronal loss or more severe (macroscopic) changes in neuronal morphology.

The MEMRI results are similar to those collected in a previous study with the commonly used pesticide, chlorpyrifos (see Hernandez et al., 2015) and indicate that an alkylphosphate nerve agent, DFP (like a phosphorothionate, insecticide OP, chlorpyrifos) can result in persistent alterations in axonal transport in the living mammalian brain. The basis for the modest increase in signal intensity associated with the 30-day washout period after the lowest dose of DFP is unclear, but may represent some type of compensatory effect. Transient increases in axonal transport (considered a form of plasticity) have been observed in CNS neurons after acute injury in both rodent and non-human primate models (Brock et al., 2018). Here it is important to note, however, that we did not observe any evidence of acute neuronal injury with the doses of DFP evaluated (especially the lowest dose) and we did not observe an increase in axonal transport after exposure to the lowest dose of chlorpyrifos in the previous (Hernandez et al., 2015) study. The method we used (MEMRI) is a non-invasive imaging approach that has gained popularity in the last few years to detect impairments of fast axonal transport in multiple contexts including aging in rats (Cross et al., 2008), transgenic mouse models of Alzheimer's disease (Kim et al., 2011; Smith et al., 2007, 2011), and frontotemporal dementia (Majid et al., 2014). In the current study (similar to the Hernandez et al., 2015 study), we utilized the method to visualize Mn^{2+} enhancement of the optic nerve projections (see Lin et al., 2014) from the retina to the superior colliculus. From intravitreal injection, Mn^{2+} has been shown to enter retinal ganglion cells and to travel within their axons in the anterograde direction along the optic nerve to the superior colliculus and lateral geniculate nucleus (Bearer et al., 2007; Watanabe et al., 2001).

While the role of a potential relationship between the myelin-related alterations we observed associated with DFP exposure and axonal transport deficits is unclear, previous studies in mice have indicated that fast axonal transport is dependent on oligodendrocyte function. For example, using a mouse model of hereditary spastic paraplegia type 2 due to a null mutation of the myelin Plp gene, a progressive impairment in fast retrograde and anterograde transport was observed (Edgar et al., 2004). Interestingly, the loss of oligodendrocyte support of myelinated axons disrupts mitochondrial function and reduces adenosine triphosphate (ATP) production leading to axonal transport deficits, axon swelling, and eventually Wallerian degeneration (Nave, 2010; Ferreirinha et al., 2004). Thus, it appears that oligodendrocyte function/myelination and the axonal transport process have mutually dependent roles in neuronal function. Future studies will be designed to investigate the effects of OPs on key components of the myelination process (e.g., levels of myelin basic protein levels, myelin proteolipid protein).

Additional mechanisms of the DFP-related alterations in axonal transport will also be the subject of future experiments. There is now a significant amount of evidence that OPs may alter the function of motor proteins such as kinesin and/or components of the neuronal cytoskeleton (e.g., microtubules) that are important for axonal transport (reviewed, Terry, 2012). For example evidence that OPs might interact with kinesin to negatively affect kinesin-driven axonal transport is supported by previous *in vitro* studies (microtubule motility assays) (Gearhart et al., 2007) as well a mass spectrometry studies where (using the biotin-tagged OP agent, FP-biotin) OP binding to tyrosine in the human kinesin 3C motor domain was demonstrated (Grigoryan et al., 2009). There is also evidence that OPs impair tubulin polymerization leading to the disruption of microtubule assembly, which in turn leads to impairments of axonal transport (Prendergast et al., 2007; Grigoryan and Lockridge, 2009). Interestingly the oxon metabolite of chlorpyrifos has also been shown to covalently bind to tubulin, an effect that may explain the disruptions in tubulin polymerization (Jiang et al., 2010). Moreover, our recent study in rats indicated that chronic exposure to chlorpyrifos leads to a significant and persistent inhibition of tubulin acetylation in the brain, another key regulator of axonal transport (Yang et al., 2018).

Other recent publications that may have relevance to the results of this study and OP effects on cytoskeletal proteins (discussed in the preceding paragraph) come from studies designed to identify peripheral biomarkers of neurotoxicity in veterans with GWI and farmers exposed to pesticides who developed neurological signs and symptoms of neural injury (Abou-Donia et al., 2017; El Rahman et al., 2018). In study 1, the serum of 20 veterans with GWI and 10 non-veteran symptomatic (low back pain) controls were screened for the presence of autoantibodies to central nervous system-specific proteins, while in study 2, serum samples from 50 farmers chronically exposed to different mixtures of pesticides (including a variety of OPs) and 25 non-exposed controls were analyzed. In each study, higher levels of autoantibody reactivity against the following proteins were detected in the veterans with GWI and pesticide exposed farmers (compared to controls): neurofilament triplet proteins (NFP), tubulin, microtubule associated tau proteins (Tau), microtubule associated protein-2 (MAP-2), myelin basic protein (MBP), myelin associated glycoprotein (MAG), glial fibrillary acidic protein (GFAP), and calcium-calmodulin kinase II (CaMKII). The findings of the two studies described above are similar to a previous study (Abou-Donia et al., 2013) of flight crew members who were anecdotally exposed to cabin air emissions containing OPs and who experienced symptoms now referred to as "Aerotoxic Syndrome". Collectively, while the human studies cited here are relatively small, retrospective, and/or descriptive, they would appear to support the argument that OP exposure may lead to the generation of autoantibodies that target proteins known to play critical roles in both the structure and function of neurons including myelination and axonal transport. An "autoimmune" response might offer one explanation for why OP exposures could lead to chronic (in some cases lifelong) symptoms.

There are some potential limitations to our studies that should be noted. While we have suggested that adverse effects on myelin and axonal transport may underlie our previous observations of DFP-related memory-related impairments, we have not specifically evaluated the 0.5 mg/kg/day x 14 days dosing approach in learning and memory-related studies previously. In older studies, we did evaluate the 0.250 mg/kg/day x 14 days dosing approach and observed prolonged spatial learning deficits as well as decreases in presynaptic cholinergic receptors (Prendergast et al., 1997; Stone et al., 2000). Moreover, we only evaluated the effects of a DFP-washout period in the vivo MEMRI studies and did not make this assessment in the brain volume or EM analyses. Thus, given the persistence of the DFP-related effects on axonal transport in vivo, it is unclear if the effects on myelin would have persisted (or even have been exacerbated) after a DFP-washout. Similarly, DFP effects on brain volume might be slow to emerge and, therefore not apparent only 24 h after the last DFP exposure. While our dosing approach was designed to model repeated exposures to OP levels not associated with overt cholinergic toxicity, the levels of exposure in the context of common agriculture exposures and long-terms effects, GWI or Aerotoxic syndrome are unknown. Moreover, it is possible that a single OP exposure (as opposed to repeated exposures) could lead to the aforementioned symptoms of these conditions. In our studies, RBC and brain AChE was inhibited by approximately 70% to 80% in RBCs and brain, respectively, by the 0.5 mg/kg dose, which has not been documented in humans who suffer from GWI or Aerotoxic syndrome. Here it should be noted that up to two weeks of repeated exposures by guinea pigs to the nerve agent sarin at 0.4 x the LD_{50} resulted in a decrease of RBC AChE activity to approximately 10% of baseline levels without any overt signs of toxicity (Shih et al., 2006). Similarly, we have observed decreases in plasma cholinesterase activity to approximately 10% of baseline levels in rats repeated exposed to chlorpyrifos

18.0 mg/kg without overt signs of toxicity (Terry et al., 2007). Collectively, these observations suggest are suggestive of some type of adaptive mechanism or tolerance to OP exposures when the doses are below the threshold of acute toxicity.

In conclusion, results of this study indicate that repeated exposures to the nerve agent, DFP at doses that are below the threshold for acute toxicity, can result in persistent decreases in axonal transport as well as alterations in myelin structure in the rodent brain. These observations could explain some of the long-term neurological deficits that have been observed in humans who have been repeatedly exposed to OPs.

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